



MERISTEM

MERITRADE PLATFORM

Production Specification Document

NGX Trading Portal Overhaul

Version: 1.0

Date: February 2026

Classification: Confidential

Meristem Securities Limited
Nigeria

DOCUMENT CONTROL

Document Information

Field	Value
Document ID	MERITRADE-PROD-SPEC-001
Version	1.0
Status	FINAL - Approved for Implementation
Classification	CONFIDENTIAL - Internal Use Only
Owner	Product Management Unit
Next Review Date	February 7, 2027
Effective Date	March 1, 2026

Revision History

Version	Date	Changes	Author	Status
1.0	2025-12-10	Initial draft	Adeyinka Freeman	Superseded

Approvals

Role	Name	Date
Chief Privacy Officer	[Signature Required]	2026-02-07
Chief Information Security Officer	[Signature Required]	2026-02-07
General Counsel	[Signature Required]	2026-02-07
Data Protection Officer (EU/DE)	[Signature Required]	2026-02-07
Chief Executive Officer	[Signature Required]	2026-02-07

Table of Contents

DOCUMENT CONTROL	2
Executive Summary	6
Document Purpose	6
Key Objectives.....	6
1. SYSTEM OVERVIEW	7
1.0 Introduction and Observations	7
1.0.1 Justification.....	7
1.1 Platform Architecture	9
1.2 Core Components.....	9
2. User Registration & Onboarding.....	11
2.1 Purpose	11
2.2 Onboarding State Machine	11
2.3 Data Classification & NDPR Compliance	12
2.4 Tokenization Strategy	13
3. Trading Engine & Order Lifecycle Management	14
3.1 Objectives.....	14
3.2 Supported Order Types	14
3.3 Order State Machine.....	14
3.4 Pre-Trade Validation.....	15
3.4.1 Account-Level Checks.....	15
3.4.2 Order-Level Checks	15
4. Market Data & NGX Connectivity	16
4.1 Data Architecture	16
4.2 Market Data Sources	16
4.2.1 Primary Sources.....	16
4.2.2 Secondary Sources	16
4.3 Live vs Delayed Data Segregation.....	16
4.4 Market Session Awareness.....	16
5. Security, Access Control & Compliance	17
5.1 Security Objectives	17
5.2 Authentication Architecture	17
5.2.1 Authentication Methods.....	17
5.3 Role-Based Access Control (RBAC)	17

5.4 Data Encryption	17
5.4.1 Encryption at Rest	17
5.4.2 Encryption in Transit.....	17
5.5 Audit Logging.....	18
6. Portfolio Management, Positions & P&L	19
6.1 Objectives.....	19
6.2 Position Model	19
6.3 P&L Calculations	19
6.3.1 Unrealized P&L	19
6.3.2 Realized P&L	19
6.4 Corporate Actions Handling	19
7. Surveillance, Risk Monitoring & Market Abuse Controls	20
7.1 Objectives.....	20
7.2 Real-Time Risk Controls	20
7.2.1 Pre-Trade Risk Checks	20
7.2.2 Post-Trade Risk Checks.....	20
7.3 Market Abuse Pattern Detection	20
7.4 Case Management & Escalation.....	20
8. Deployment Architecture & Operations.....	22
8.1 Environment Segregation	22
8.2 CI/CD Pipeline	22
8.3 Monitoring & Observability	22
8.3.1 Metrics	22
8.3.2 Structured Logging	22
8.3.3 Distributed Tracing	22
8.4 Disaster Recovery	23
9. Testing & Quality Assurance Strategy	24
9.1 Testing Objectives	24
9.2 Testing Framework	24
9.3 Test Coverage Requirements	24
10. Compliance & Regulatory Framework	25
10.1 Regulatory Alignment.....	25
10.2 Reporting Requirements	25
10.3 Audit & Evidence	25

Appendices	26
Appendix A: API Reference	26
A.1 Authentication & User Management.....	26
Appendix B: Data Dictionary	26
Appendix C: Glossary	27
Appendix D: Integration Points	28
Appendix F – PROTOTYPE.....	29
Document Control.....	40

Executive Summary

This document provides a comprehensive production specification for the complete overhaul of the Meritrade platform, Meristem Securities Limited's flagship trading desk and platform for the Nigerian Exchange (NGX).

The platform redesign encompasses end-to-end trading operations, from user registration through trade execution and settlement, with strict adherence to Nigerian Data Protection Regulation (NDPR), Securities and Exchange Commission (SEC) requirements, and NGX connectivity standards.

Document Purpose

This specification serves as the authoritative technical and functional blueprint for:

- Development teams implementing the platform
- Quality assurance and testing teams
- Compliance and regulatory stakeholders
- Operations and support teams
- External auditors and regulatory bodies

Key Objectives

- Full NDPR compliance with tokenization and data protection
- Seamless NGX connectivity for real-time trading
- Multi-channel support (Web and Mobile) with feature parity
- End-to-end audit trail and regulatory reporting
- Market-grade performance and reliability
- Comprehensive surveillance and risk controls

1. SYSTEM OVERVIEW

1.0 Introduction and Observations

The Meritrade platform is a web-based/mobile app platform for trading and investing in Nigerian shares. The portal also includes a platform for portfolio management, stock recommendations, and a "WealthMakers" club.

1.0.1 Justification

Overhauling the existing platform are key requirements by existing customers. The issues identified in the Meritrade platform are not cosmetic or isolated usability defects; they indicate structural weaknesses across technology architecture, operational processes, and service governance. Unattended, these limitations materially impact user experience, regulatory compliance, and the platform's competitiveness within Nigeria's evolving digital capital markets ecosystem. Key aspects identified include: -

a. System Stability and Performance Risk

- Persistent application crashes, session instability, and forced re-authentication point to deficiencies in:
- Session state management
- Backend scalability under concurrent load
- Inefficient synchronization between market data, portfolio state, and UI rendering

In a real-time trading environment, such instability creates execution risk, information asymmetry, and potential client detriment, which exposes the platform to regulatory scrutiny and reputational damage.

b. Authentication and Access Control Weaknesses

- Recurring login failures and frequent forced logouts suggest:
- Weak session token lifecycle management
- Poor handling of device-level authentication states
- Insufficient resilience against network latency or mobile OS interruptions

This directly undermines:

- User confidence in account security
- Platform usability for active traders

- Compliance with best-practice access control standards expected by regulators and institutional partners

c. KYC and Withdrawal Friction

- Reported difficulties in fund withdrawals and repeated KYC requests indicate:
- Fragmented or poorly integrated identity verification workflows
- Lack of a clear, deterministic KYC state machine
- Inadequate separation between regulatory verification, operational controls, and user experience
- Such friction increases:
- Customer attrition risk
- Perceived opacity around fund safety
- Regulatory exposure under AML/CFT and consumer protection expectations

d. Support and Operational Responsiveness Gaps

- Slow or unresponsive customer support reflects:
- Absence of structured case management
- Weak escalation and SLA enforcement
- Limited observability into user-reported system failures

For a financial services platform, this is not merely a service issue; it constitutes an operational risk, particularly during market volatility or settlement-related disputes.

e. Functional and Strategic Limitations

- Compared to international trading platforms, Meritrade's feature set lacks:
- Advanced analytics and charting
- Intelligent alerts and decision support
- Seamless portfolio insight and reporting
- Robust audit-grade transparency for user actions

This constrains:

- Engagement of sophisticated retail and high-net-worth users
- Platform differentiation in an increasingly competitive fintech landscape
- Long-term scalability into advisory, wealth management, or institutional segments

f. Strategic Imperative

Given:

- Rising expectations of Nigerian retail investors
- Increased regulatory emphasis on system resilience, auditability, and consumer protection
- Growing competition from global and regional digital trading platforms

There is a clear justification for a structured redesign and enhancement of the Meritrade platform, focusing on:

- Architectural resilience
- Secure, seamless user journeys (web and mobile)
- Deterministic KYC and settlement workflows
- Feature depth aligned with modern trading standards
- Strong governance, auditability, and operational controls

This transformation is necessary not only to remediate current deficiencies but to position the platform as a credible, scalable, and compliant digital brokerage solution within the Nigerian capital market ecosystem.

1.1 Platform Architecture

The Meritrade platform is a comprehensive multi-tier trading system designed for the Nigerian capital markets. The architecture comprises the following layers:

- **Presentation Layer:** Web application and native mobile applications (iOS/Android)
- **Application Layer:** Business logic, trading engine, portfolio management, and notification services
- **Integration Layer:** NGX connectivity, bank integrations, CSCS interfaces, and external data providers
- **Data Layer:** Tokenized user data vault, market data store, audit logs, and operational databases
- **Infrastructure Layer:** Cloud hosting, security services, monitoring, and disaster recovery

1.2 Core Components

The platform consists of 33 major components organized into functional domains:

Components	Domain	Description
1-4	User Management	Registration, KYC, Authentication, User Profiles

5-8	Trading Operations	Order Management, Execution, Market Data, Portfolio
9-12	Risk & Compliance	Surveillance, Audit, Reporting, Compliance
13-16	Data Management	Market Data, Analytics, Notifications, Alerts
17-21	Infrastructure	Security, Deployment, Monitoring, DR, Testing
22-26	Integration	NGX, Banks, CSCS, Corporate Actions, TradingView
27-33	Operations	Back Office, Settlement, Support, Admin, Education

2. User Registration & Onboarding

2.1 Purpose

Enable prospective users to create accounts, prove identity, link to trading profiles, and achieve a legally compliant state where trading is permitted.

Hard Constraint: No trade-related UI is accessible until onboarding state = APPROVED.

2.2 Onboarding State Machine

The platform tracks onboarding as an immutable state machine, ensuring auditability and preventing inconsistent states:

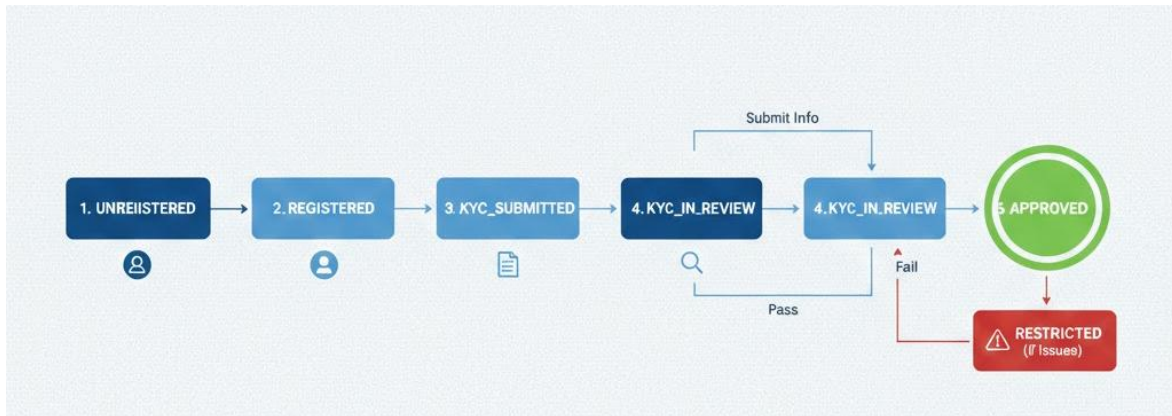


Fig.1.1.1. Onboarding High Level Process Flow

State	Description	Trigger Event
UNREGISTERED	User has not created an account	User lands on platform
REGISTERED	Account created, credentials verified	Email/phone verified via OTP
KYC_SUBMITTED	Identity documents uploaded	BVN, NIN, and ID documents submitted
KYC_IN_REVIEW	Compliance team reviewing KYC	Automated and manual verification in progress
APPROVED	KYC approved, trading enabled	All checks passed, trading UI unlocked
REJECTED	KYC failed, resubmission allowed	User can correct and resubmit documentation
RESTRICTED	Account temporarily suspended	Triggered by surveillance or compliance events

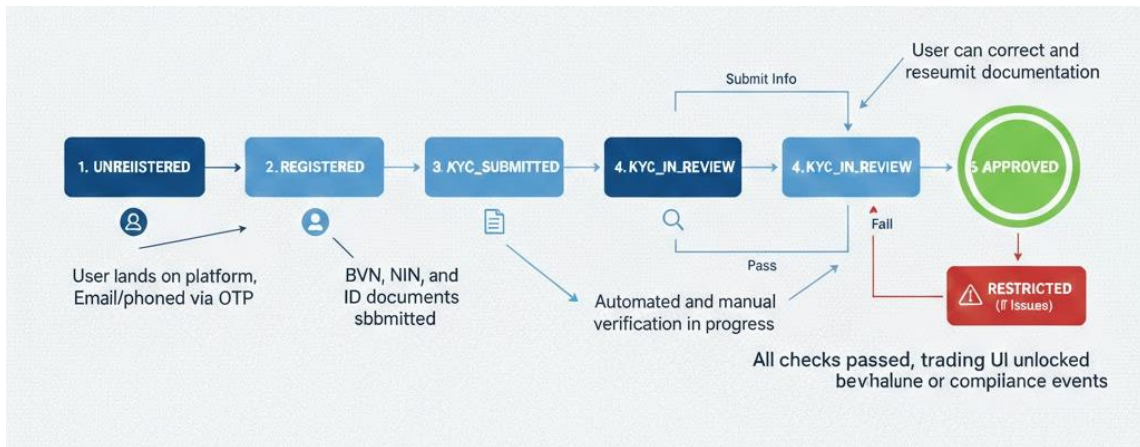


Fig 2: Business process Flow – Onboarding

2.3 Data Classification & NDPR Compliance

All personal data is classified into three protection domains to ensure NDPR compliance:

Domain	Type	Examples	Protection Rules
Domain A	Direct Identifiers (High Risk)	Name, DOB, Address, BVN, NIN	Never stored in plaintext; tokenized immediately
Domain B	Quasi-Identifiers	CSCS, Bank account, Phone, Email	Tokenized immediately; displayed only in masked form
Domain C	Operational Identifiers	USER_UUID, TRADING_ACCOUNT_UUID	Safe for all trading operations and logs

2.4 Tokenization Strategy

At data ingestion, the system implements the following tokenization flow:

1. User submits PII (BVN, NIN, bank account) via Web or Mobile
2. Tokenization Service generates randomized, non-reversible tokens
3. Raw PII encrypted and stored in isolated vault
4. Application layer receives only tokens; raw PII never exposed
5. Vault access requires compliance authorization and is fully audited

3. Trading Engine & Order Lifecycle Management

3.1 Objectives

The trading engine ensures exchange-validated trades without ambiguity, race conditions, or audit gaps. This component is subject to the highest regulatory scrutiny.

3.2 Supported Order Types

Order Type	Description	Session Availability	Usage Level
Market	Execute immediately at best available price	All sessions	High
Limit	Execute only at specified price or better	All sessions	Medium
Stop	Trigger market order when stop price reached	Open session only	Medium
Stop-Limit	Trigger limit order when stop price reached	Open session only	Low
Day	Valid until end of trading day	All	Standard
IOC	Immediate-Or-Cancel (fill immediately or cancel)	Open session	Low
FOK	Fill-Or-Kill (fill completely or cancel)	If NGX supports	Low

3.3 Order State Machine

All orders transition through immutable, event-driven states:

State	Description	Trigger Event
NEW	Order received by system	User submits order via UI
VALIDATED	Pre-trade checks passed	Risk, account, and order validation complete
SENT	Forwarded to NGX	Order transmitted via trading gateway
ACKNOWLEDGED	NGX confirms receipt	Exchange acknowledgment received
PARTIALLY_FILLED	Partial execution	Part of order quantity executed
FILLED	Fully executed	Complete order quantity matched
CANCELLED	Order cancelled	User or system cancellation confirmed
REJECTED	Order rejected	Validation failure or exchange rejection

EXPIRED	Order expired	Time-in-force period elapsed
----------------	---------------	------------------------------

3.4 Pre-Trade Validation

Before any order reaches NGX, the system performs comprehensive validation:

3.4.1 Account-Level Checks

- KYC status = APPROVED
- Account active and not restricted
- Trading entitlement verified
- Risk limits not breached

3.4.2 Order-Level Checks

- Symbol tradable on NGX
- Market session state allows order type
- Price within market bounds (circuit breakers)
- Quantity respects minimum lot size
- Sufficient buying power available

Critical Rule: If validation fails, the order is rejected locally and never forwarded to NGX.

4. Market Data & NGX Connectivity

4.1 Data Architecture

The market data layer transforms raw NGX feeds into reliable, queryable market intelligence while maintaining exchange truth and audit capability.

4.2 Market Data Sources

4.2.1 Primary Sources

- NGX Trading Engine (orders, trades, execution confirmations)
- NGX Market Data Feed (real-time prices, volumes, order book depth)
- Corporate Actions Feed (dividends, splits, rights issues)
- Instrument Master Feed (symbol metadata, ISINs, tick/lot sizes)

4.2.2 Secondary Sources

- Historical price vendors
- FX reference rates
- Sector and index classifications

Critical Rule: No crowd-sourced or scraped data is permitted. All market data must originate from authorized sources.

4.3 Live vs Delayed Data Segregation

Data Type	Usage	Access Control	Delivery	Caching
Live Data	Trade execution, order entry, portfolio management	Entitlement-controlled	WebSocket streaming	Not cached long-term
Delayed Data	Analytics, public market views, historical charts	15-minute delay	REST API with caching	Aggressive caching allowed

4.4 Market Session Awareness

The data layer maintains awareness of market session states and gates functionality accordingly:

- **Pre-Open:** Order entry allowed, no execution, indicative pricing only
- **Open:** Full trading, real-time execution, order book visibility
- **Closing Auction:** Limited order types, auction matching rules
- **Halt:** Trading suspended, no new orders, cancellations allowed
- **Closed:** No trading activity, market data frozen at close

5. Security, Access Control & Compliance

5.1 Security Objectives

The security layer guarantees confidentiality, integrity, availability, non-repudiation, and regulatory compliance. This is not optional—without this layer, all trading and analytics capabilities are non-deployable.

5.2 Authentication Architecture

5.2.1 Authentication Methods

- Username + password (hashed with bcrypt, salted)
- Multi-Factor Authentication (MFA): SMS/Email OTP, TOTP authenticator apps, optional hardware tokens
- Biometric login (mobile): FaceID, TouchID, Android Biometrics
- OAuth 2.0 / OpenID Connect (enterprise SSO)
- API keys (machine-to-machine access)

5.3 Role-Based Access Control (RBAC)

Role	Capabilities	Scope
End User	Trade execution, portfolio view, alert management	Own account only
Trader	Advanced trading, bulk orders, limit management	Designated accounts
Portfolio Manager	Portfolio analytics, performance reporting	Managed portfolios
Compliance Officer	Surveillance, audit logs, KYC review	Read-only system-wide
Support Agent	User assistance, case management	Read-only positions/trades
System Admin	Infrastructure, role management, configuration	Full system access (logged)
External Auditor	Tokenized reports, time-limited access	Compliance-approved scope

5.4 Data Encryption

5.4.1 Encryption at Rest

- AES-256 encryption for all databases and backups
- Separate encryption keys per environment
- PII stored exclusively in secure vault with additional encryption layer

5.4.2 Encryption in Transit

- TLS 1.3 for all external connections
- Mutual TLS for internal service-to-service communication
- Certificate pinning for mobile applications

5.5 Audit Logging

Every security-relevant event is captured in immutable audit logs:

- Authentication events (login, logout, MFA, failed attempts)
- Permission denials and privilege escalation attempts
- Trade execution and order lifecycle events
- Alert generation and report access
- PII vault access (with purpose and authorization tracking)
- Configuration changes and system events

Retention: Minimum 7 years for regulatory compliance, with cold storage for long-term archival.

6. Portfolio Management, Positions & P&L

6.1 Objectives

Maintain accurate positions at all times, correctly compute realized and unrealized P&L, handle partial fills and corporate actions, and support audit-grade reconstruction of portfolio history.

6.2 Position Model

Positions are tracked per account and per instrument with the following core fields:

Field	Description	Notes
Account Token	Tokenized USER_UUID and TRADING_ACCOUNT_UUID	NDPR-compliant identifier
Instrument ID	NGX symbol or ISIN	Links to instrument master
Quantity	Number of shares held	Updated on each trade
Average Cost	Weighted average purchase price	Recalculated on buys
Market Value	Current price × quantity	Real-time update
Unrealized P&L	(Market price - Avg cost) × quantity	Mark-to-market
Realized P&L	Cumulative profit/loss from closed positions	Immutable once recorded
Open/Closed Status	Position state indicator	Event-driven transition

6.3 P&L Calculations

6.3.1 Unrealized P&L

Formula: $(\text{Current Market Price} - \text{Average Cost}) \times \text{Open Quantity}$

Updated in near real-time as market prices change.

6.3.2 Realized P&L

Formula: $(\text{Sell Price} - \text{Average Cost}) \times \text{Quantity Sold}$

Recorded at trade execution and immutable thereafter.

6.4 Corporate Actions Handling

The system processes corporate actions with position and cash adjustments:

- **Dividends:** Cash credited to account, recorded in cash ledger
- **Stock Dividends:** Quantity adjusted, average cost recalculated
- **Bonus Issues:** Quantity increased proportionally, average cost adjusted
- **Stock Splits:** Quantity multiplied, average cost divided
- **Rights Issues:** Entitlement recorded, exercise tracked separately
- **Consolidations:** Reverse split, quantity reduced, average cost increased

Critical Rule: All corporate action processing is effective-date driven and fully logged. Manual overrides require compliance approval.

7. Surveillance, Risk Monitoring & Market Abuse Controls

7.1 Objectives

Detect abnormal or abusive trading behavior, enforce risk limits in real-time, support post-trade forensic analysis, and produce defensible regulatory evidence. For NGX-connected brokers, surveillance is a licensing condition.

7.2 Real-Time Risk Controls

7.2.1 Pre-Trade Risk Checks

- Maximum order size per transaction
- Maximum order value limits
- Daily trading volume limits per user
- Symbol concentration limits (% of portfolio)
- Buying power and margin validation

7.2.2 Post-Trade Risk Checks

- Position limit breaches
- Rapid turnover pattern flags
- Loss threshold monitoring
- Margin utilization tracking

7.3 Market Abuse Pattern Detection

Pattern	Description	Detection Method
Spoofing	Large orders placed then cancelled to manipulate price	Pattern analysis: order size vs cancel frequency
Layering	Multiple orders across price levels to create false liquidity	Time-window analysis: simultaneous multi-level orders
Wash Trades	Self-trading across accounts to inflate volume	Cross-account correlation and matching
Momentum Ignition	Coordinated aggressive orders to trigger price movement	Velocity analysis: rapid order placement patterns
Insider Trading	Trading ahead of material non-public information	Pre-announcement volume spikes, unusual account activity

7.4 Case Management & Escalation

Each flagged incident becomes a case with:

- Unique case ID and creation timestamp
- Linked orders, trades, and market context
- Timeline reconstruction and pattern analysis
- Analyst notes and investigation findings
- Resolution outcome and regulatory reporting status

Critical Rule: Cases are immutable once closed. All escalations and resolutions are fully audited.

8. Deployment Architecture & Operations

8.1 Environment Segregation

Environment	Purpose	NGX Access	Data Type
Development (DEV)	Feature development and unit testing	No	Synthetic/anonymized
Integration (INT)	System integration testing	No	Synthetic/anonymized
UAT	Business and compliance validation	No	Anonymized production-like
Production (PROD)	Live trading operations	Yes	Real user data (tokenized)

Hard Rule: No environment except PROD has live NGX connectivity.

8.2 CI/CD Pipeline

The deployment pipeline enforces quality gates at each stage:

6. **Code Commit:** Developer commits trigger automated build
7. **Static Analysis:** Security scans, code quality checks, dependency audits
8. **Unit Tests:** ≥90% code coverage requirement
9. **Integration Tests:** Component interaction validation
10. **Build Artifact:** Immutable, versioned container images
11. **INT Deployment:** Automated deployment to integration environment
12. **Automated E2E Tests:** Full workflow validation
13. **UAT Deployment:** Business and compliance validation
14. **Approval Gates:** Engineering Lead, Product Owner, Compliance Officer
15. **Production Release:** Canary deployment with progressive rollout

8.3 Monitoring & Observability

8.3.1 Metrics

- Order submission latency (target: <300ms)
- Trade execution latency (target: <200ms)
- Market data refresh rate (target: <500ms)
- Alert trigger latency (target: <2s)
- Order rejection rate, API response times, system throughput

8.3.2 Structured Logging

All logs are JSON-formatted with correlation IDs, timestamp synchronization, and NDPR-compliant pseudonymization.

8.3.3 Distributed Tracing

Every critical user action generates a trace ID that follows the request through all services:

TRACE_ID → UI → API Gateway → Trading Engine → NGX → Response

8.4 Disaster Recovery

Metric	Target	Recovery Method
RPO (Recovery Point Objective)	≤ 5 minutes	Continuous replication, encrypted backups
RTO (Recovery Time Objective)	≤ 30 minutes	Hot standby, automated failover
Backup Frequency	Real-time	Cross-region replication
DR Testing	Quarterly	Documented failover procedures, drill verification

9. Testing & Quality Assurance Strategy

9.1 Testing Objectives

Validate all features before production, ensure zero data loss or corruption, maintain regulatory-grade audit trails, and enable rapid rollback for failed deployments.

9.2 Testing Framework

Test Type	Scope	Execution
Unit Tests	Individual modules (order engine, alerts, KYC)	Automated on every commit, ≥90% coverage
Integration Tests	Component interactions (trades ↔ alerts ↔ reporting)	Simulated NGX feed and bank integration
End-to-End Tests	Complete user journeys (registration → settlement)	Multiple user types, cross-device validation
Performance Tests	Market surge simulations, high-frequency trading stress	Load testing with realistic volumes
Security Tests	Vulnerability scanning, penetration testing	Periodic external audits, automated scanning

9.3 Test Coverage Requirements

- **Unit Tests:** ≥90% code coverage for critical components
- **Integration Tests:** All NGX, bank, and CSCS integration points
- **E2E Tests:** Complete user journeys from registration to settlement
- **Performance Tests:** Market surge scenarios, high-frequency trading stress
- **Security Tests:** Penetration testing, vulnerability scanning, compliance audits

10. Compliance & Regulatory Framework

10.1 Regulatory Alignment

The platform aligns with the following regulatory requirements:

- **NDPR (Nigeria Data Protection Regulation):** Data minimization, tokenization, consent management, and purpose limitation
- **SEC Nigeria:** Capital market regulations, reporting requirements, and best execution standards
- **NGX Rules:** Connectivity standards, market data usage, and trading member obligations
- **CSCS:** Central Securities Clearing System integration and settlement protocols
- **CBN (Central Bank of Nigeria):** Payment system integration and fund transfer regulations

10.2 Reporting Requirements

Report Type	Frequency	Recipient	Content
Trade Blotter	Daily	NGX, SEC	All executed trades, timestamps, order details
Position Reports	Daily	CSCS	End-of-day positions, settlement instructions
Surveillance Reports	As triggered	SEC, Compliance	Suspicious activity cases, patterns
KYC Status	Monthly	Internal Compliance	Onboarding states, approval rates
System Uptime	Monthly	NGX, Management	Availability metrics, incident reports

10.3 Audit & Evidence

All compliance activities generate immutable evidence:

- Timestamped audit logs with cryptographic signatures
- Immutable trade blotters and order lifecycle records
- KYC documentation with consent tracking
- Surveillance case files with investigation timelines
- Regulatory submission receipts and acknowledgments

Appendices

Appendix A: API Reference

The platform exposes RESTful APIs and WebSocket endpoints for client applications and internal services.

A.1 Authentication & User Management

Endpoint	Method	Purpose	Auth Required
POST /api/v1/user/register	POST	Create new user account	No
POST /api/v1/user/login	POST	Authenticate user	No
POST /api/v1/user/mfa/verify	POST	Verify MFA token	Session
GET /api/v1/user/profile	GET	Retrieve user profile	JWT
POST /api/v1/kyc/submit	POST	Submit KYC documents	JWT
GET /api/v1/kyc/status	GET	Check KYC status	JWT
POST /api/v1/orders/place	POST	Submit trade order	JWT + MFA
GET /api/v1/orders/{id}	GET	Get order status	JWT
DELETE /api/v1/orders/{id}	DELETE	Cancel order	JWT + MFA
GET /api/v1/portfolio	GET	Retrieve portfolio	JWT
WS /api/v1/market/stream	WebSocket	Real-time market data	JWT

Appendix B: Data Dictionary

Core data entities and their attributes:

Entity	Key Attributes	Storage Location
User	USER_UUID, email_token, phone_token, kyc_status	User database (tokenized)
Trading Account	TRADING_ACCOUNT_UUID, USER_UUID, CSCS_token, bank_token	Trading database
Order	order_id, TRADING_ACCOUNT_UUID, symbol, type, quantity, price, state	Order database
Trade	trade_id, order_id, execution_price, quantity, timestamp, ngx_ref	Trade database
Position	TRADING_ACCOUNT_UUID, instrument_id, quantity, avg_cost, unrealized_pnl	Portfolio database

Audit Log	event_id, USER_UUID, event_type, timestamp, metadata	Audit database (immutable)
------------------	--	-------------------------------

Appendix C: Glossary

Term	Definition
BVN	Bank Verification Number - unique identifier for Nigerian bank customers
CSCS	Central Securities Clearing System - settlement and clearing infrastructure for NGX
DMA	Direct Market Access - electronic trading platform for institutional investors
KYC	Know Your Customer - identity verification and due diligence process
NDPR	Nigeria Data Protection Regulation - data privacy law governing personal information
NGX	Nigerian Exchange Group - stock exchange operating in Nigeria
NIN	National Identification Number - unique identifier for Nigerian citizens
OMS	Order Management System - software for routing and managing trade orders
P&L	Profit and Loss - financial performance measurement
RBAC	Role-Based Access Control - security model assigning permissions by role
SEC	Securities and Exchange Commission - regulatory authority for capital markets
TLS	Transport Layer Security - cryptographic protocol for secure communications
UUID	Universally Unique Identifier - unique tokenized reference

Appendix D: Integration Points

External system integration specifications:

System	Protocol	Purpose	Redundancy
NGX Trading Engine	FIX 4.4 / Proprietary API	Order submission, execution confirmations	Dual connectivity
NGX Market Data	WebSocket / Multicast	Real-time prices, order book, trades	Hot standby
CSCS	REST API	Settlement instructions, position confirmation	Automated reconciliation
NIBSS BVN	SOAP / REST	BVN verification, bio-data retrieval	Retry with backoff
Bank APIs	REST / ISO 20022	Account verification, fund transfers	Alternate endpoints
TradingView	Charting Widget API	Interactive charting and technical analysis	CDN failover

Appendix F – PROTOTYPE

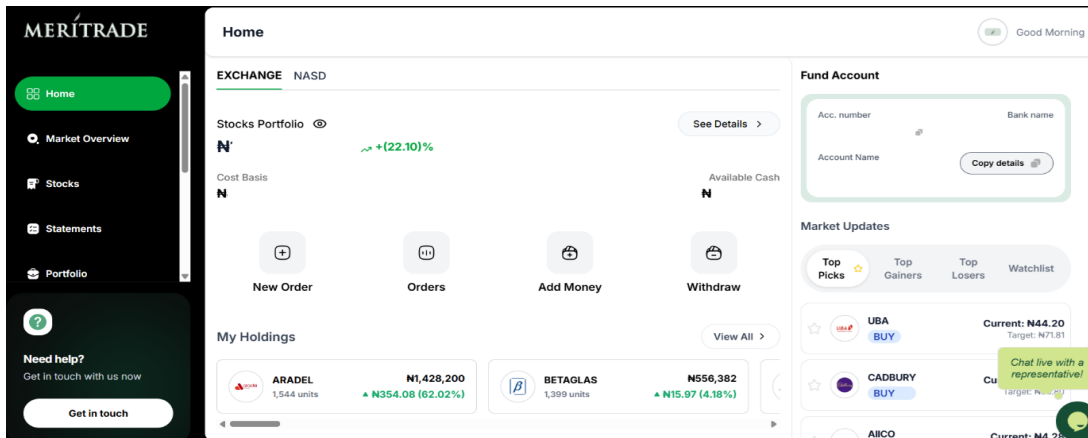


Diagram 1: Screenshot – LOGIN PAGE (AS-IS)

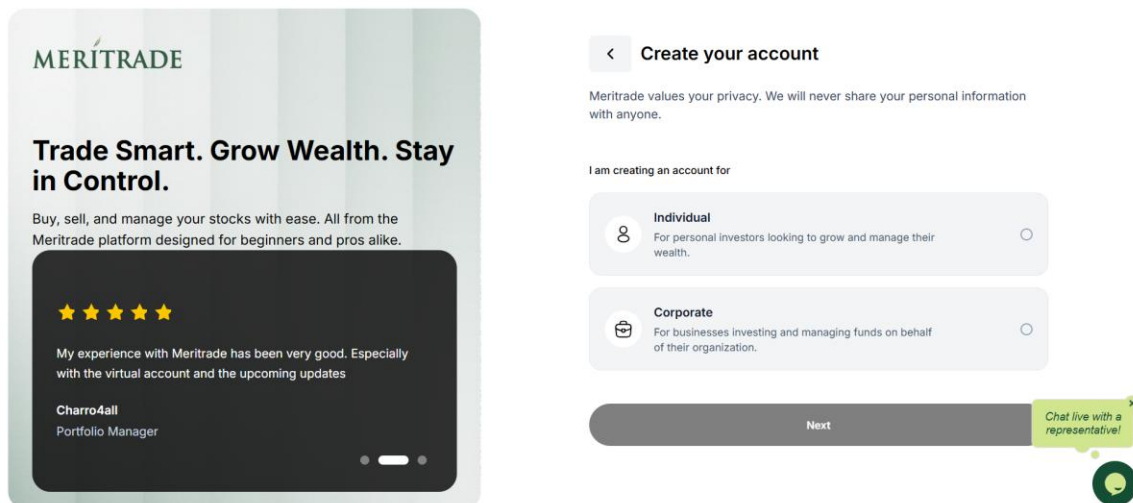


Diagram 2: Screenshot – REGISTRATION PAGE (AS-IS)

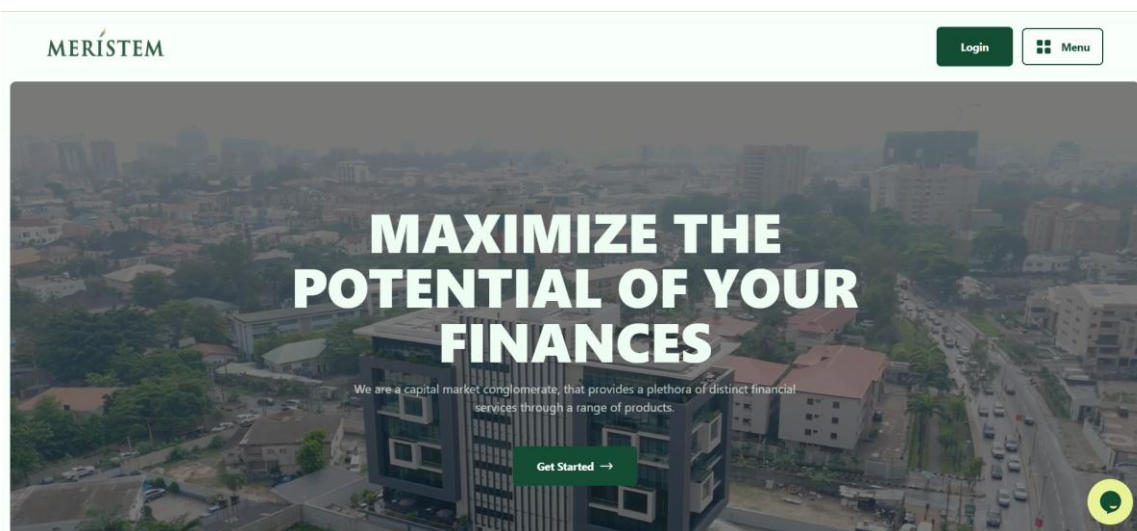


Diagram 3: Screenshot – LANDING PAGE (AS-IS)

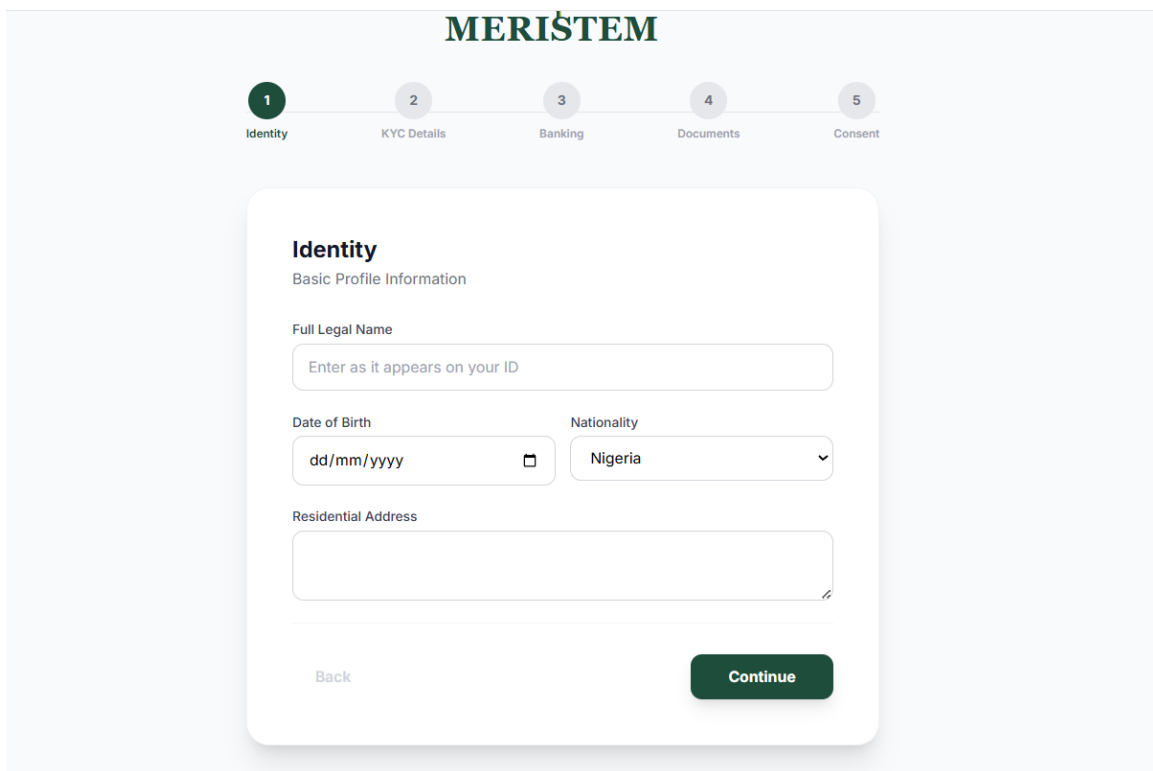
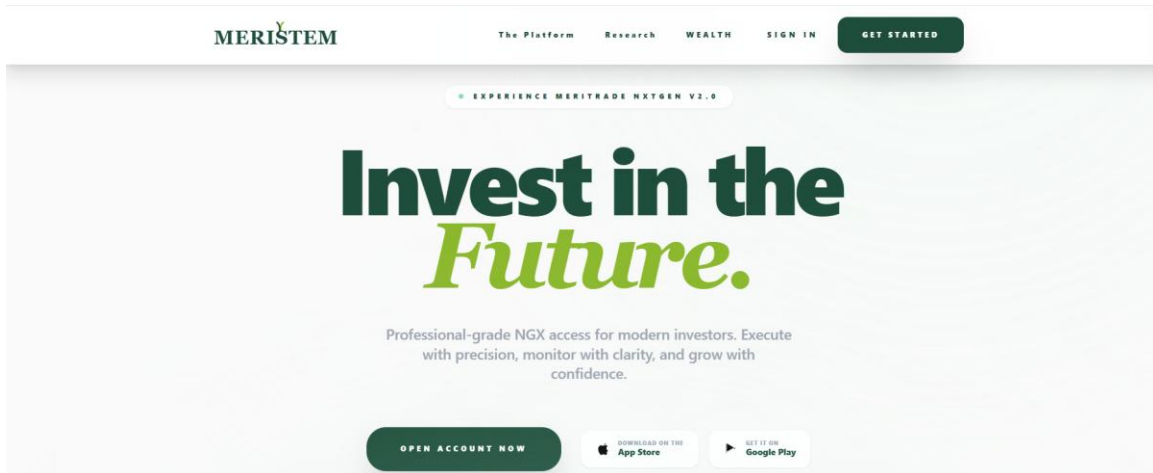


Diagram 4: Screenshot – LANDING PAGE (Proposed TO-BE)

✓

Identity

✓

KYC Details

✓

Banking

✓

Documents

5

Consent

Consent

Risk Disclosure & Privacy

RISK DISCLOSURE AGREEMENT

Trading in securities involves significant risks. The value of investments and the income from them can fall as well as rise. You may not get back the amount you originally invested...

Meristem Nigeria Limited is regulated by the Securities and Exchange Commission (SEC) and is a member of the Nigerian Exchange Group (NGX)...

DATA PROCESSING POLICY

In accordance with the Nigeria Data Protection Act (NDPA), your personal identifiers (PII) are stored in an encrypted isolated vault. Meristem only utilizes tokenized references for

☐ I ACCEPT THE RISK DISCLOSURE & TERMS OF USE

☐ CONSENT TO NDPA-COMPLIANT DATA PROCESSING

Back

Submit KYC

Meristem

Meritrade NxtGEN V1.0

Device

Onboarding Incomplete

Dashboard

Good Morning, John
Welcome to your Meritrade NextGen terminal.

KYC PENDING

Total Portfolio Value
N12,450,230.00
+N100,000 (0.8%)

Available Cash
N2,340,500.00
+N500,000 (21.8%)

Unrealized P/L
N845,600.00
+N15,000 (1.8%)

Market Day Status
OPEN
Closing in 4h 12m

Top Market Gainers (NGX)

SYMBOL	LAST PRICE	CHANGE	VOLUME
DANGCEM Dangote Cement PLC	N701.00	+5.40 (0.78%)	1.3M
MTNN MTN Nigeria Comm...	N235.50	-2.10 (-0.88%)	4.5M
ZENITHBANK Zenith Bank PLC	N42.15	+0.25 (0.6%)	12.0M
GTCCO	N51.20	+1.10 (2.19%)	8.9M

View All

AI Market Insight

••NGX EQUITY MARKET SUMMARY: DANGCEM, MTNN, ZENITHBANK••

••1. Dangote Cement PLC (DANGCEM)••

••Market Sentiment:•• ••Bullish/Stable. As the NGX's most capitalized stock, DANGCEM continues to benefit from its dominant market share and the recurring Share Buyback Program, which provides a technical floor for the share price. Investor confidence remains high due to the company's ability to pass inflationary costs to consumers through strategic pricing.

••Risk Assessment:•• High concentration risk due to low free float; susceptibility to energy price volatility (Natural Gas

Diagram 5: Screenshot – PRE-KYC CONFIRMATION PAGE (Proposed TO-BE)

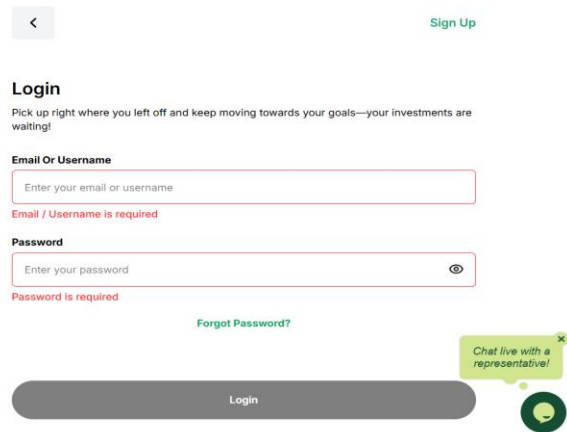
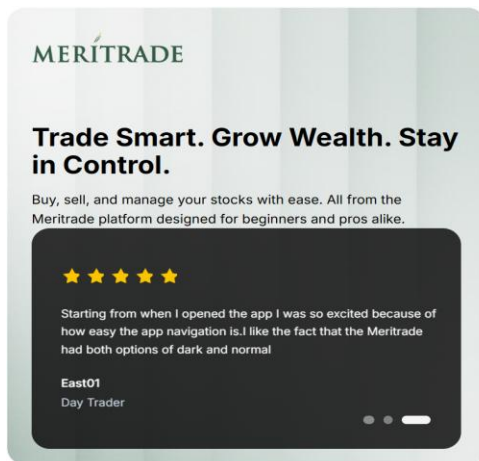


Diagram 6: Screenshot – LOGIN PAGE (AS-IS)

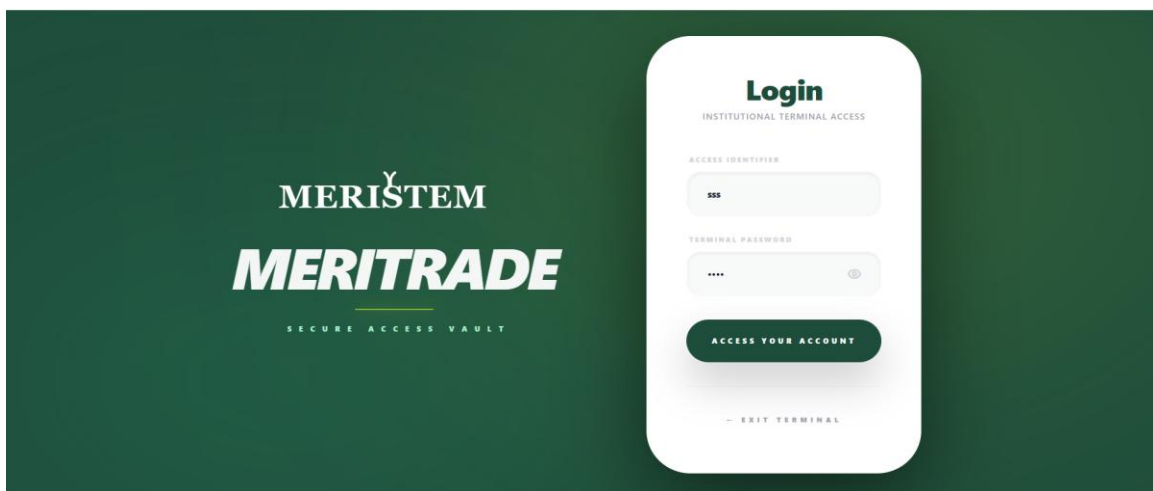


Diagram 7: Screenshot – LOGIN PAGE (TO-BE)

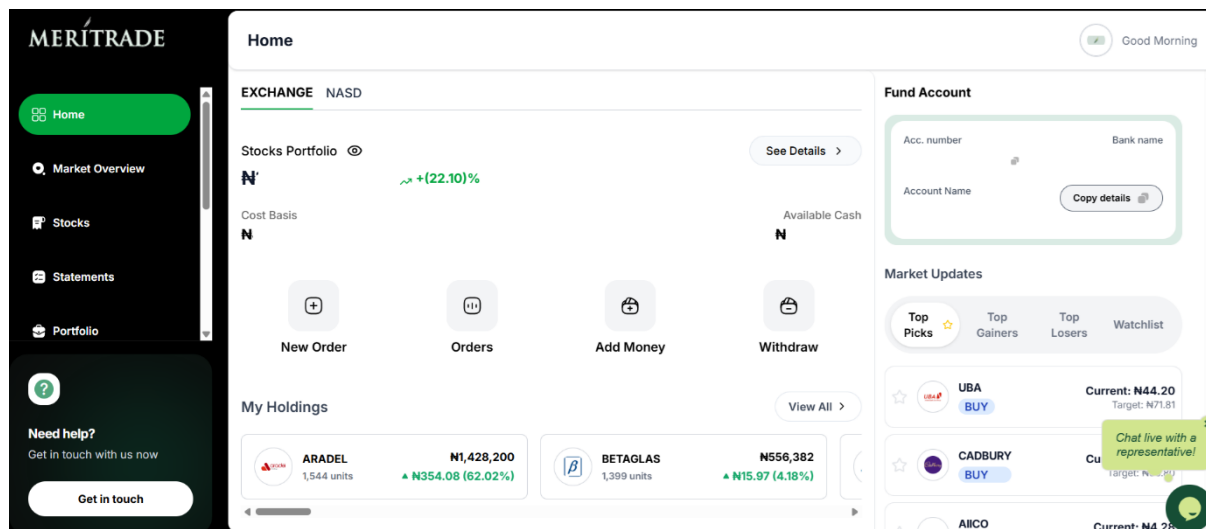


Diagram 8: Screenshot – Dashboard (AS-IS)

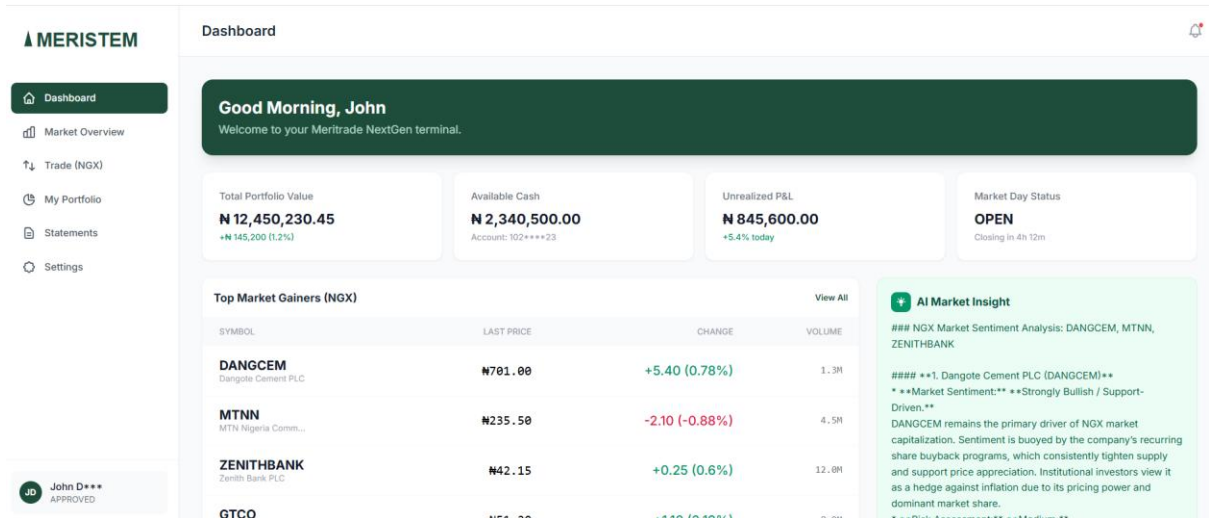


Diagram 9: Screenshot – Dashboard (TO-BE)

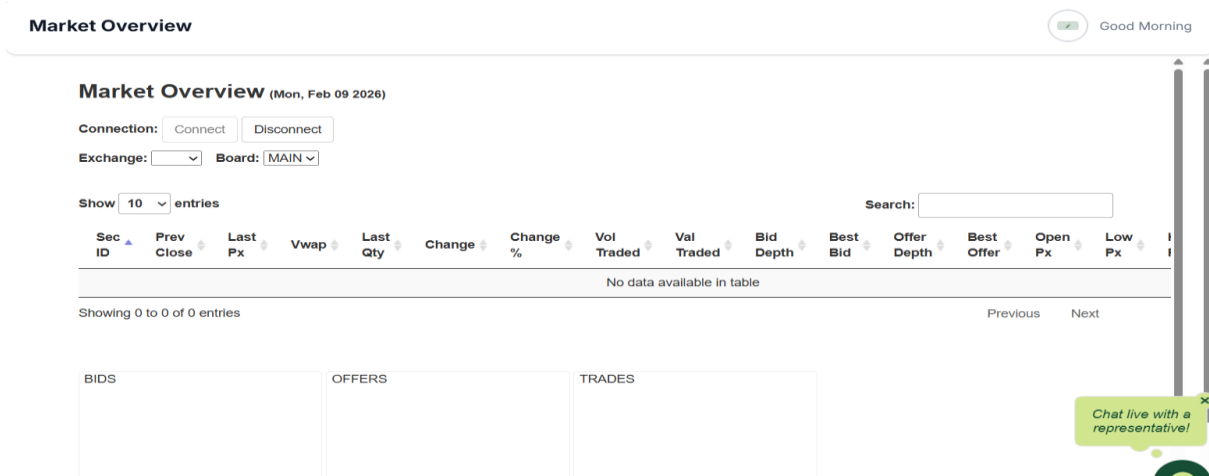


Diagram 9: Screenshot – Market Overview (As - Is)

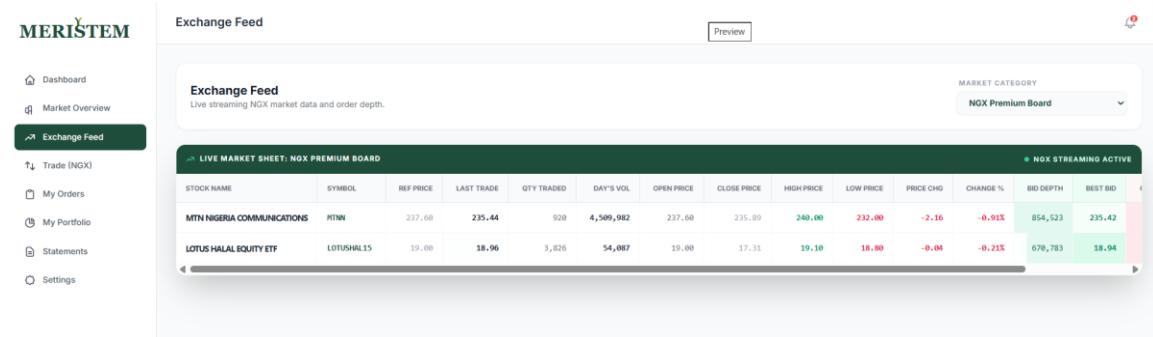


Diagram 10: Screenshot – Market Overview (To - Be)

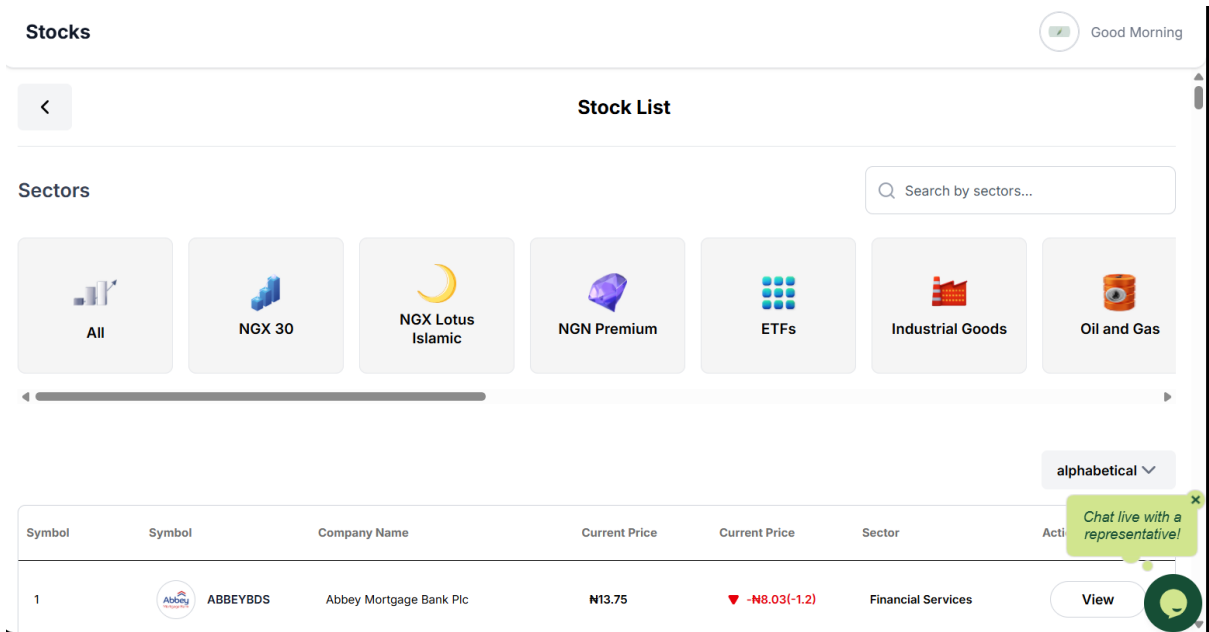


Diagram 11: Screenshot – Stock list/filter (As - Is)

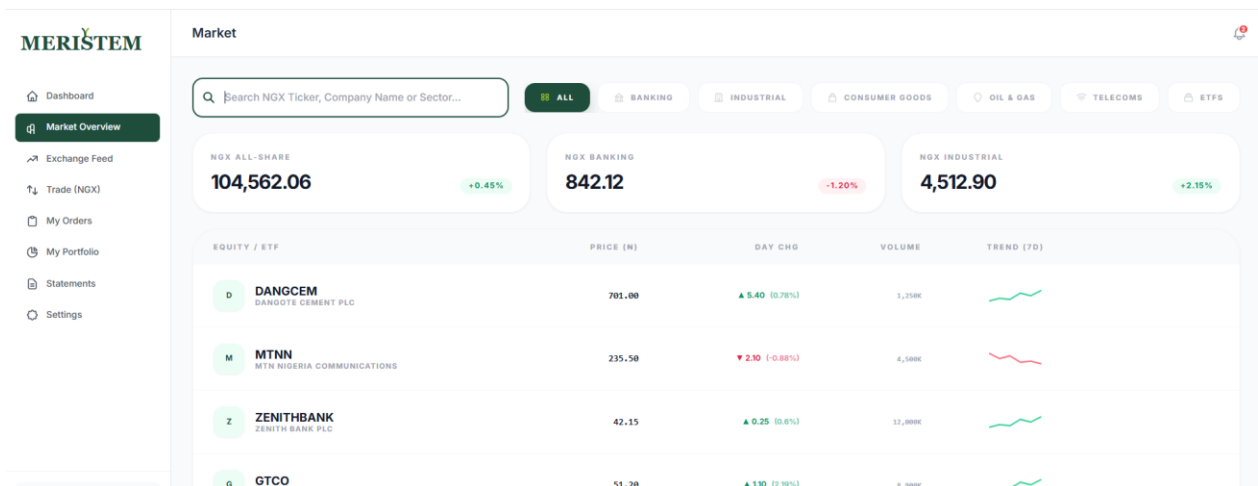


Diagram 12: Screenshot – Stock list/filter (To - Be)

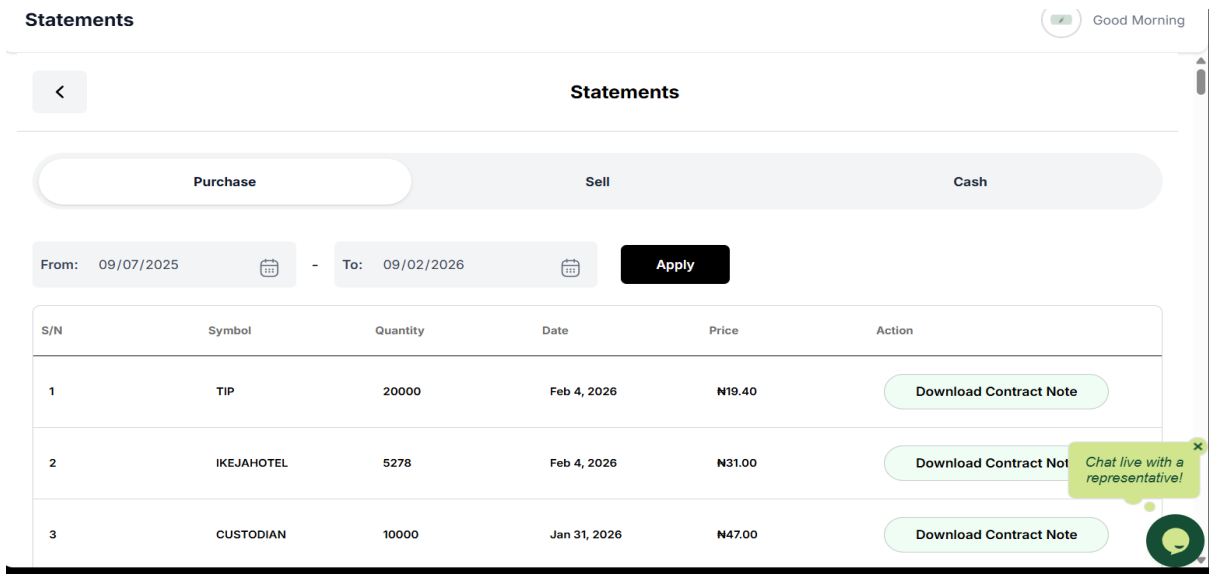


Diagram 13: Screenshot – Statement (As - Is)

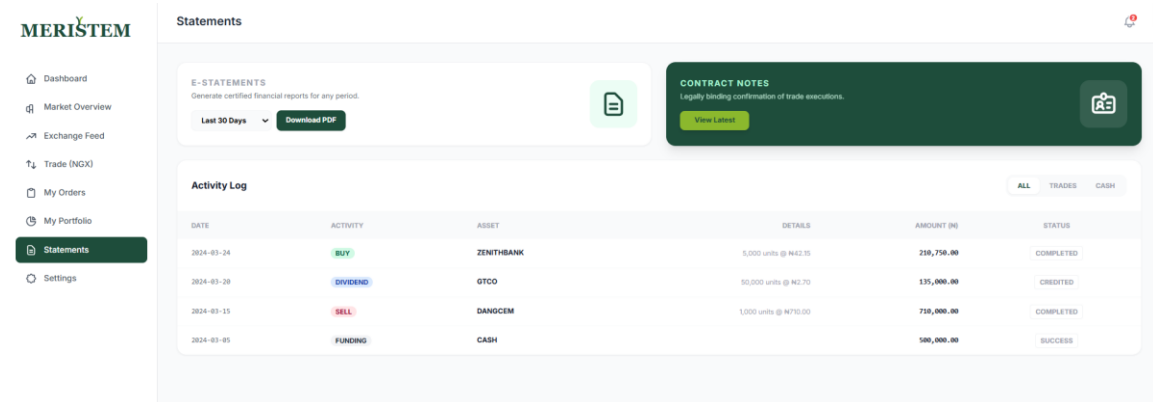


Diagram 14: Screenshot – Statement (To - Be)

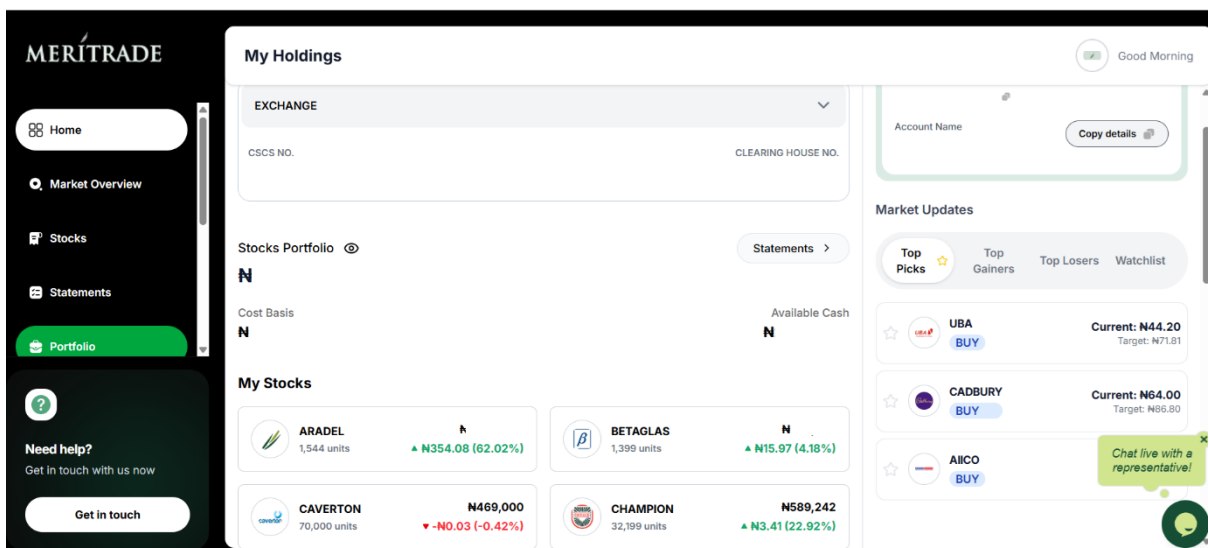


Diagram 15: Screenshot – My Holdings (As - Is)

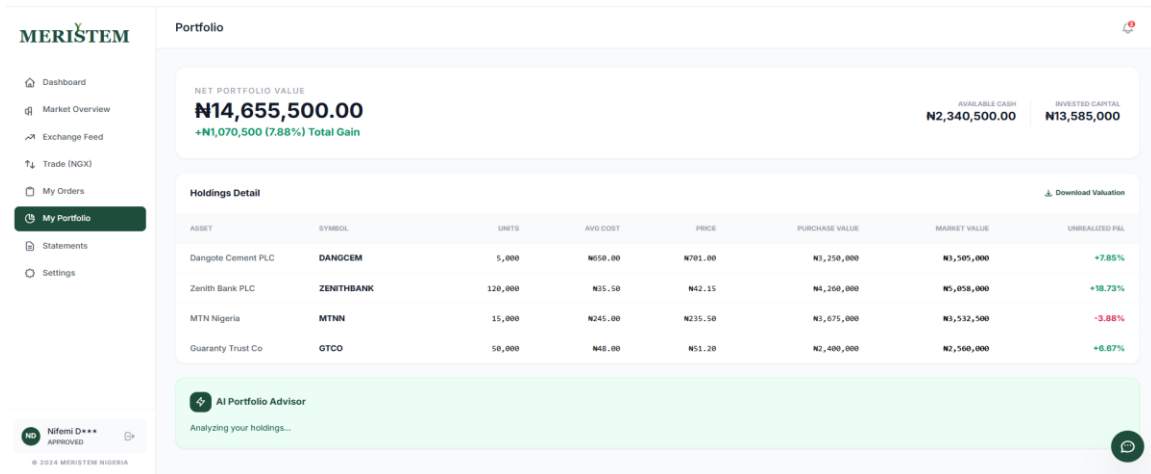


Diagram 16: Screenshot – My Holdings (To - Be)

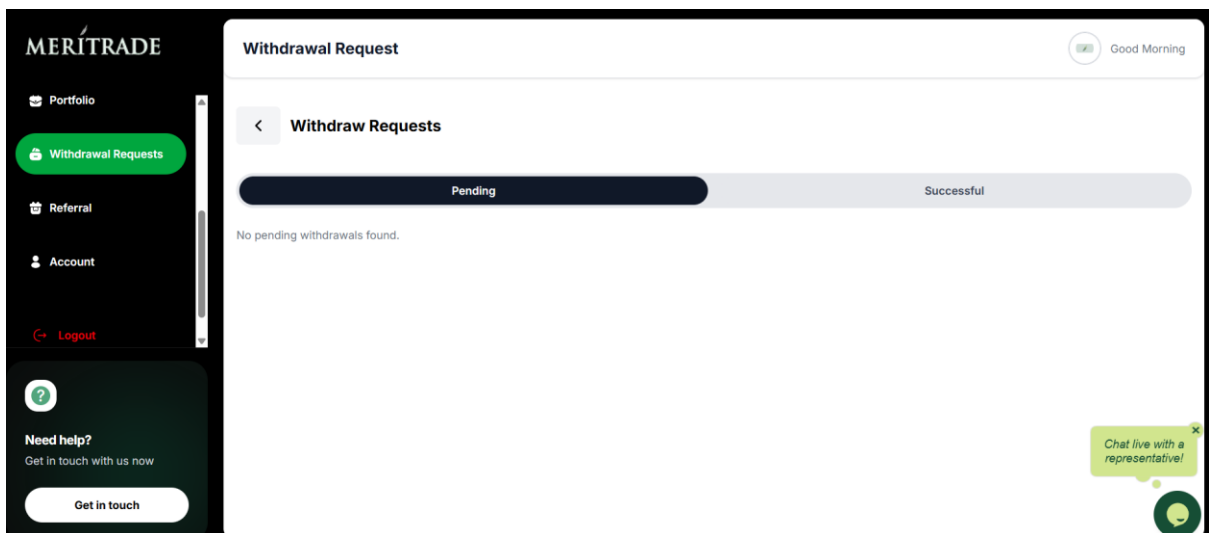
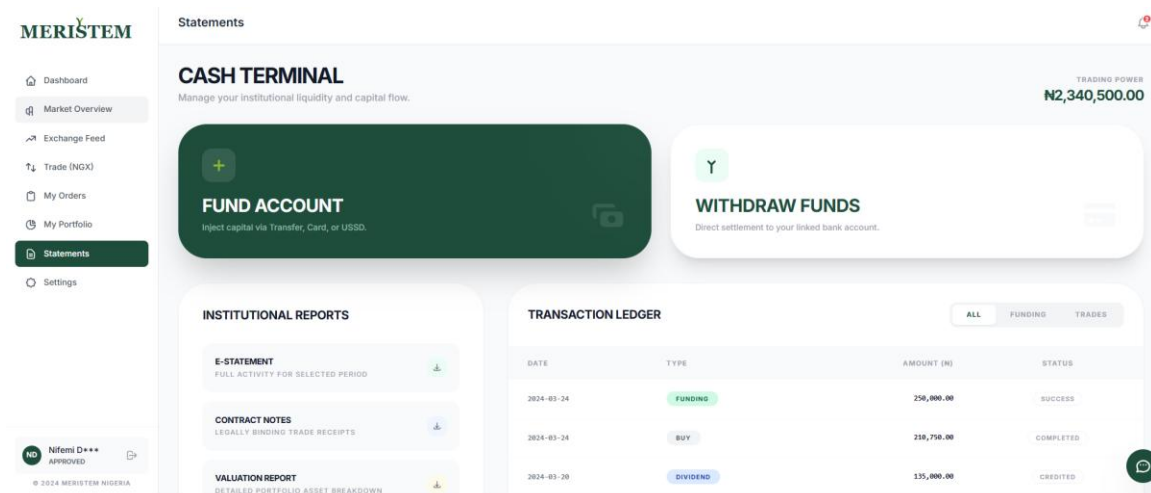


Diagram 17: Screenshot – Withdrawal request Page (As - Is)



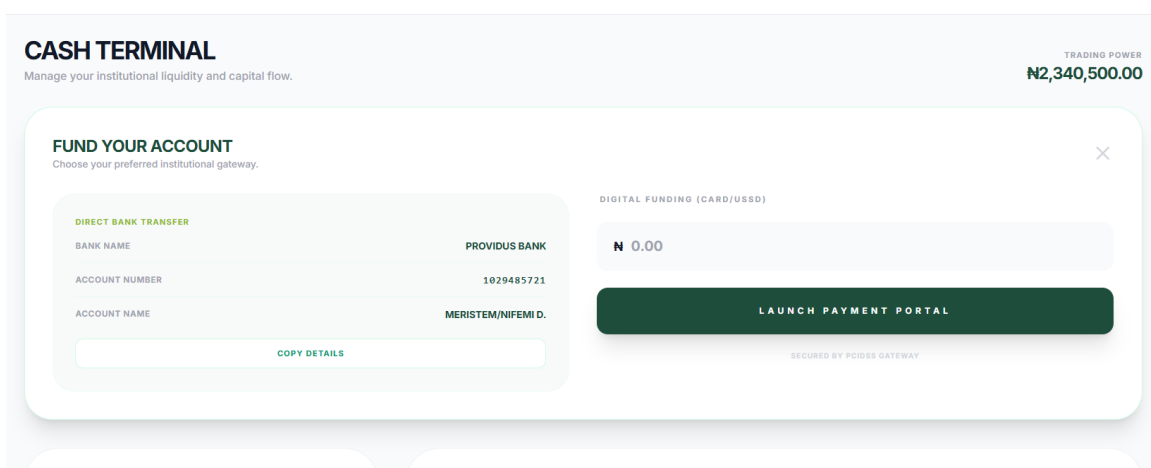
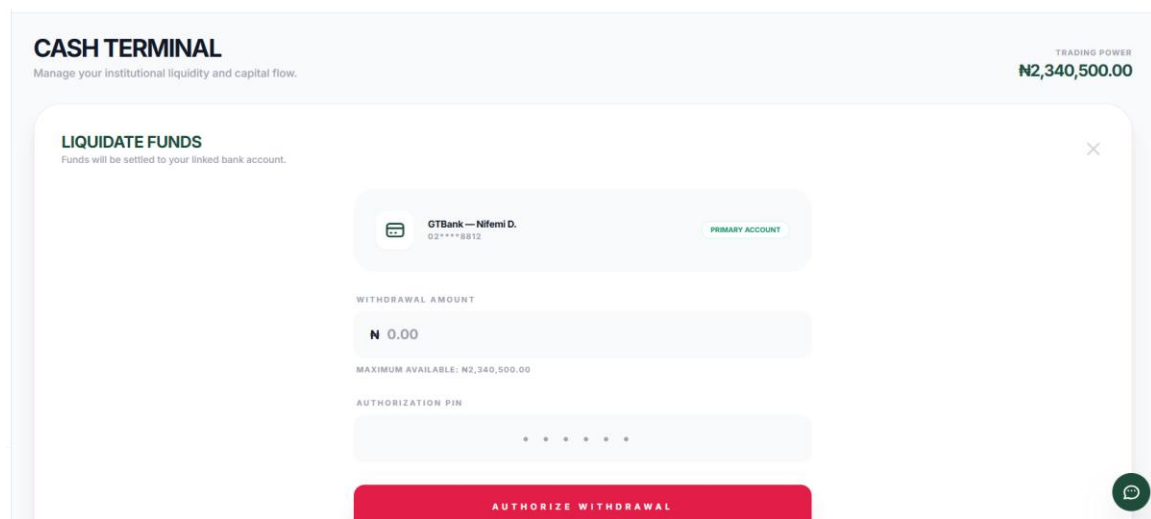


Diagram 18: Screenshot – Fund, Withdrawal request and Statement Page (To - Be)



Diagram 19: Screenshot – Trade Page (To - Be)

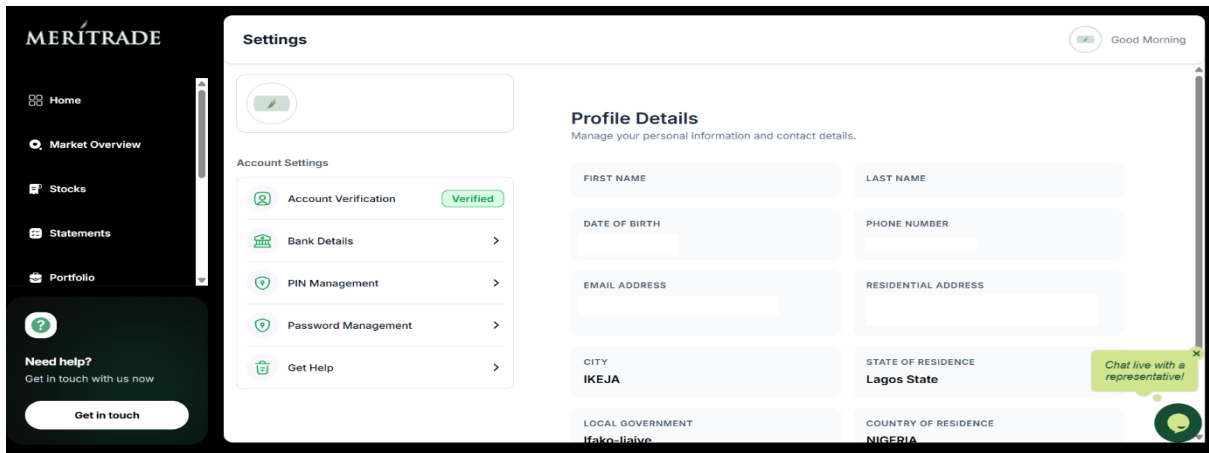


Diagram 20: Screenshot – Setting Page (As - Is)

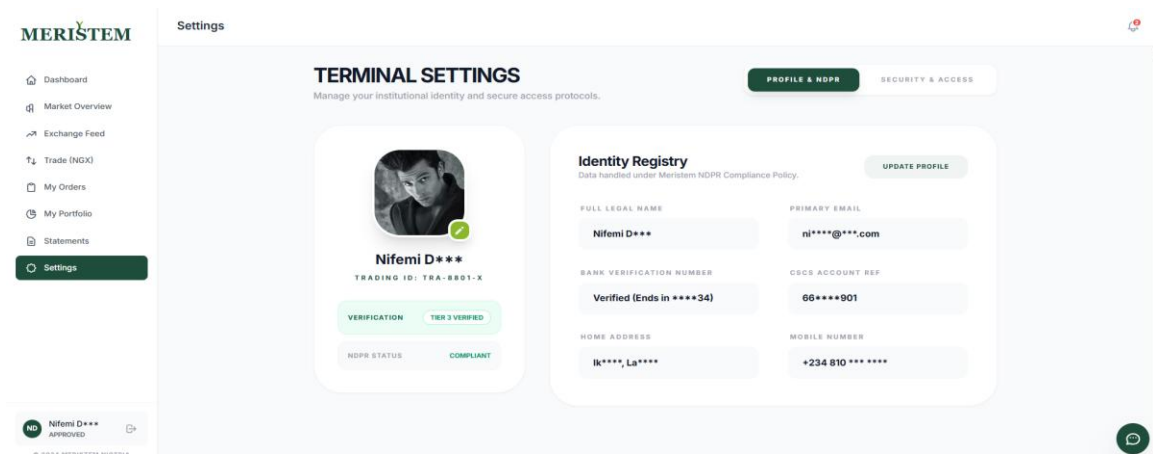


Diagram 21: Screenshot – Setting Page (To - Be)

Terminal Ready, Nifemi D.

Institutional NGX node active in Lagos, Nigeria.

TOTAL PORTFOLIO VALUE

N12,450,230.45

+N 145,200 (1.2%)

AVAILABLE CASH

N2,340,500.00

ACCOUNT: 102****23

UNREALIZED P&L

N845,600.00

+5.4% TODAY

MARKET DAY STATUS

OPEN

CLOSING IN 4H 12M

TOP GAINERS (NGX)

VIEW ALL

SYMBOL PRICE CHANGE VOLUME

DANGCEM
DANGOTE CEMENT
PLC

N701.00

+5.40%

Terminal Ready,
Nifemi D.Institutional NGX node active in Lagos,
Nigeria.

TOTAL PORTFOLIO VALUE

N12,450,230.45

+N 145,200 (1.2%)

AVAILABLE CASH

N2,340,500.00

ACCOUNT: 102****23

UNREALIZED P&L

N845,600.00

+5.4% TODAY

Diagram 22: Mobile Response – Tablet/Mobile

Document Control

Version: 1.0

Date: February 2026

Author: Meristem Securities Limited - Technology Department

Classification: Confidential - Internal Use Only

— End of Document —